Block 4 – Cloud Computing

S2: Cloud Computing & Cloud Architecture

Arquitectura de Xarxes

Universitat Pompeu Fabra

Outline

If We Can Virtualize Everything... Why Not Sell It?

Virtualization is the technology.
Cloud computing is the business model.

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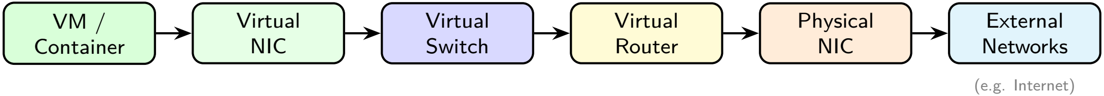
4. Identify essential cloud network services: load balancers, proxies, and VPNs

Recap: What We Virtualized (Session 1)



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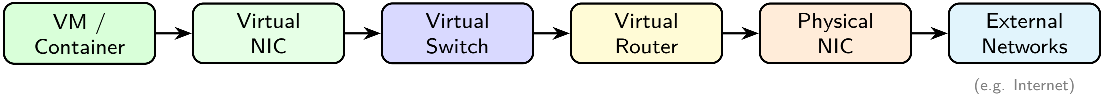

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VM / Container → Virtual NIC → Virtual Switch → Virtual Router → Physical NIC → External Networks

(e.g. Internet)

- **Compute**: VMs share physical hardware via hypervisors; containers share the host kernel directly
- **Networking**: Virtual NICs, Switches, and Routers replicate the physical stack in software
- **Isolation**: VLANs and overlay networks separate tenants on shared infrastructure

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Meet the Scenario: A Startup from Zero

Running example for this session

To understand concepts in the cloud, we will use "**CloudBite**". This is a 3-person startup that we just created and received seed funding. We need to deploy a food-delivery web app for thousands of users **by next month**.
We have laptops, code, and zero infrastructure.

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*"A model for enabling ubiquitous, convenient, **on-demand network access** to a **shared pool** of configurable computing resources that can be rapidly provisioned and released with **minimal management effort**."*

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Three phrases to remember:

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- **Shared pool**: underlying hardware is shared across many customers (multi-tenancy)
- **Minimal effort**: you click a button (or call an API); the provider handles the rest

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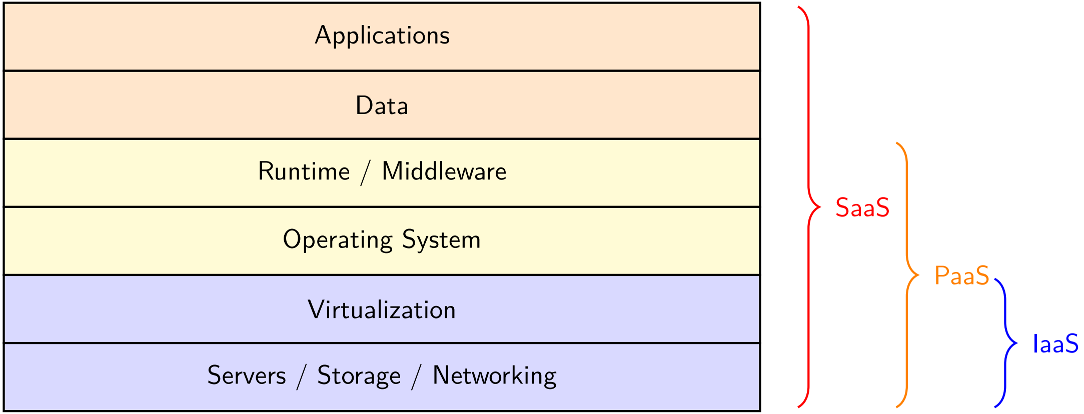
Rapid elasticity	Scale up/down automatically	Black Friday: $10\times$ more instances
Measured service	Pay only for what you use	Billed per CPU-hour, GB stored

All five must be present to qualify as "cloud computing"

Cloud Service Models – The Stack



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SaaS (e.g. Google Workspace, Microsoft 365, Slack, Zoom):

- Provider gives you a complete application. You manage your data only
- "I need email for 500 employees. I do not want to run a mail server"

Service Models: Who Manages What?

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Applications	You	You	Provider

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Network Services (LB, Proxy, VPN)	Provider	Provider	Provider

IaaS → SaaS: **less control, less responsibility**.

Major providers (AWS, Azure, GCP) offer all three models.

Discussion: Cloud Fundamentals

Case A: University

A large university wants a web tool for students to submit assignments. They have a consolidated IT department. Student data must stay within EU jurisdiction (GPDR compliant). Budget is fixed yearly.

Which model? Why?

- Control over data location needed
- Existing IT team can manage servers
- Ready-made products exist (Canvas, Moodle)
- No need to scale suddenly

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Case B: CloudBite

We are 3 engineers. We need to run a web app, store orders, and send email notifications. We must ship in 4 weeks and cannot afford to manage servers.

Which model? Why?

- No ops bandwidth: avoid IaaS if possible
- Web app: PaaS (just deploy the code)
- Email: SaaS (Mailchimp, SendGrid)
- Database: PaaS managed DB, or IaaS if we need fine-grained control

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Platform	Description	Logo
OpenStack	Open-source, widely used in private clouds	openstack
VMware vSphere	Enterprise leader, proprietary	vmware
Proxmox VE	Open-source, gaining traction since recent VMware price rises	PROXMOX

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Provider	**Description**	**Logo**
Amazon Web Services (AWS)	Market leader since 2006	aws
Microsoft Azure	Strong enterprise integration	Azure
Google Cloud Platform (GCP)	Data and AI focus	Google Cloud

These map directly to the IaaS model we just covered: the provider (or you) manages compute, storage, and networking.

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Cooling / Power	Provider	You

Public cloud trades control for simplicity; private cloud gives full control at the cost of operational overhead.

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Hosted / Public: same idea, managed by the cloud provider.

Examples: AWS ECS/EKS, Azure AKS, Google Kubernetes Engine (GKE), OpenShift on major clouds.

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- What if traffic spikes overnight?

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Key: under the hood, cloud providers use **overlay networks** (covered in Block 5) to isolate tenants at scale.

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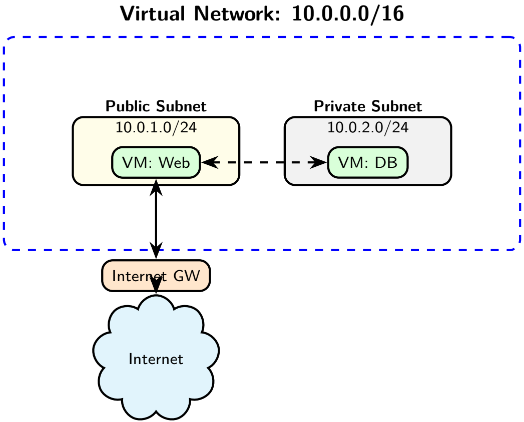
Virtual Network Architecture



Subnets: Public vs Private

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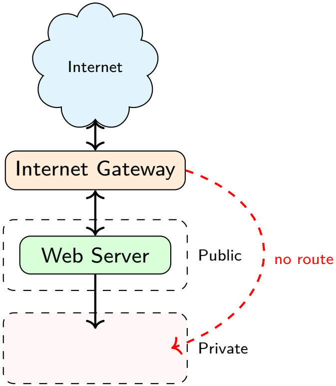
- Route to an **Internet Gateway**

Private subnet:



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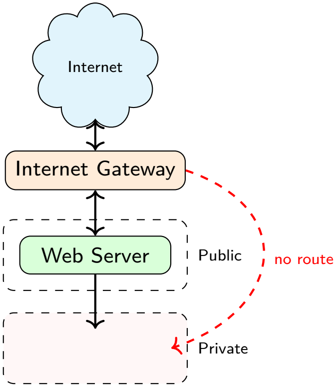

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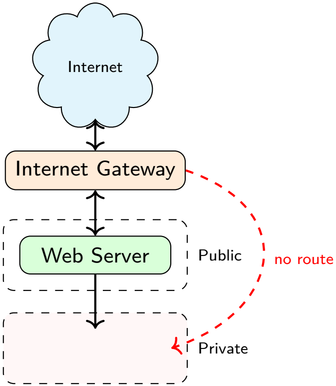

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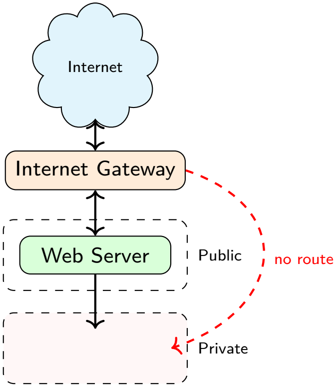

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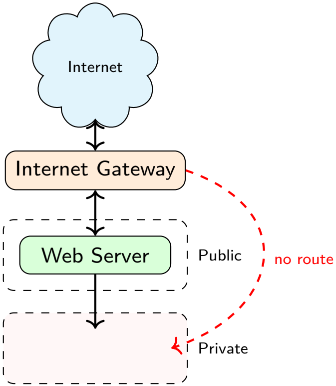

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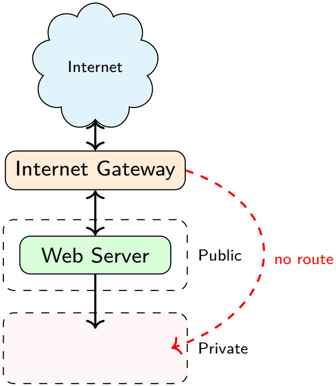

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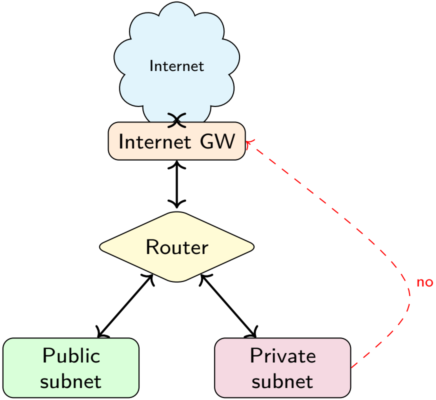

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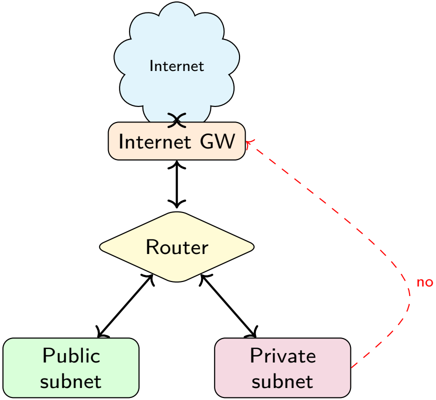

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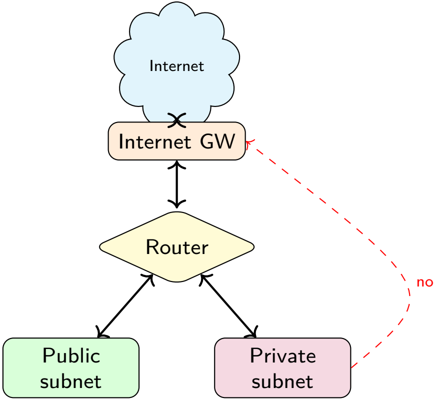

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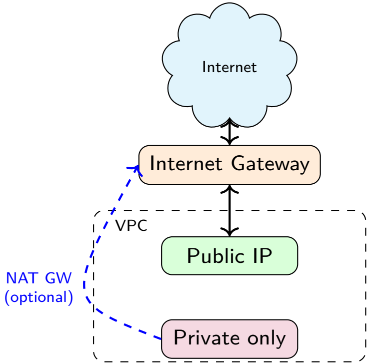

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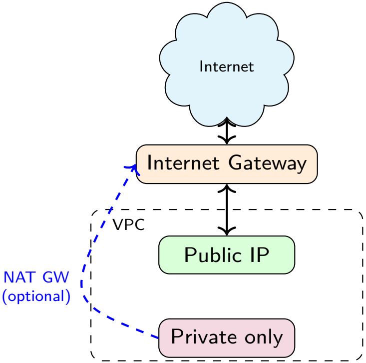

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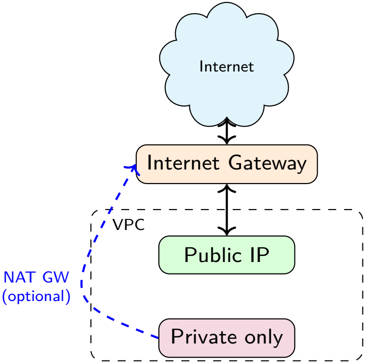

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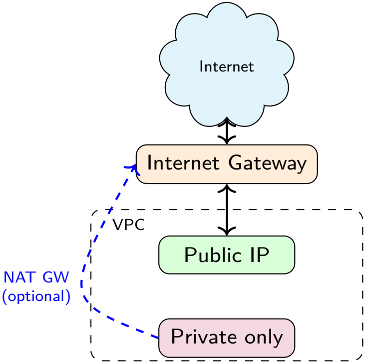

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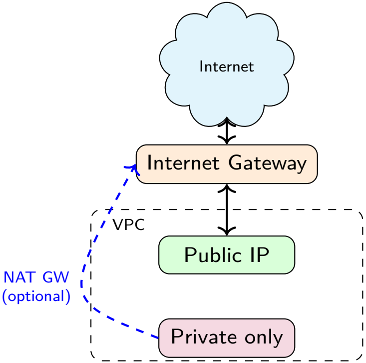

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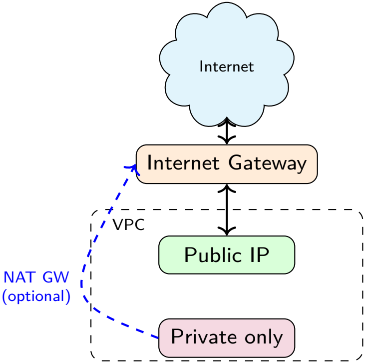

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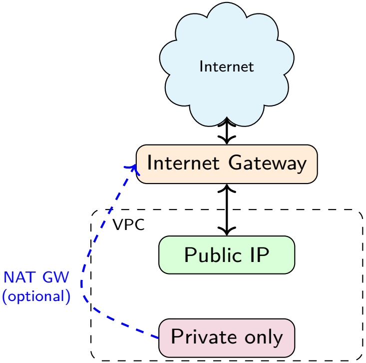

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Note: a NAT Gateway can be added for private instances that need outbound Internet access (e.g. updates), without exposing them inbound.

Security Groups: Instance-Level Firewall

Once traffic reaches an instance, you still need to control what it can send and receive.

- **Security group** = virtual firewall on an instance
- Specifies allowed **inbound/outbound** traffic
- **Stateful**: allow inbound → response automatically allowed

Dir.	Proto	Port	Src/Dst
In	TCP	80	0.0.0.0/0
In	TCP	443	0.0.0.0/0
In	TCP	22	10.0.0.0/16
Out	All	All	0.0.0.0/0

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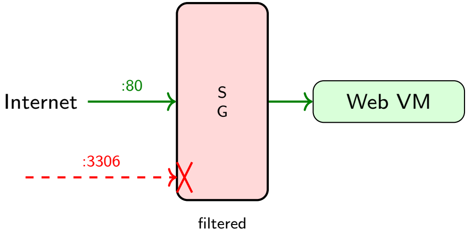


By default: all inbound denied, all outbound allowed.

[3] AWS, "Security Groups for Your VPC," docs.aws.amazon.com.

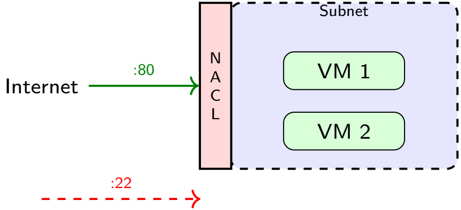
Network ACLs (Access Control Lists): Subnet-Level Firewall

Network ACLs add a layer of control at the **subnet boundary**, before traffic reaches any instance.

- **Network ACL** = firewall at the **subnet** level
- Rules evaluated in **order** (numbered, first match wins)
- **Stateless**: must explicitly allow inbound AND outbound

#	Dir.	Proto	Port	Action
100	In	TCP	80	ALLOW
200	In	TCP	443	ALLOW
*	In	All	All	DENY

4



Filter applies to **all** instances in the subnet.

[4] AWS, "Network ACLs," docs.aws.amazon.com.

Security Groups vs Network ACLs

	Security Groups	Network ACLs
Scope	Instance level	Subnet level
State	Stateful	Stateless
Rules	Allow only	Allow + Deny
Evaluation	All rules evaluated	First match wins
Default	Deny all inbound	Allow all

Defense in depth: use both together.

- Security Groups → fine-grained per-instance control
- Network ACLs → broad subnet-level guardrails

Discussion: Cloud Networking

We have running CloudBite on AWS. We need to make sure that our web server is reachable from the users, but the database is not be exposed to the Internet. How do we set this up?

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- NAT Gateway if the DB needs outbound access (e.g. pulling updates)

Outline

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Three services solve these problems: load balancers, proxies, and VPNs.

Cloud Network Services: Overview

Beyond virtual networks and security controls, public cloud providers offer managed network services that simplify common infrastructure tasks, such as:

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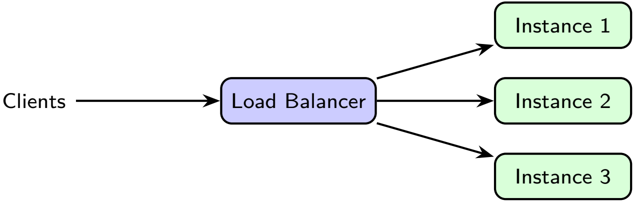
These services run as **managed offerings**: no servers to install or maintain.

Note: Proxy, Reverse Proxy, and VPN will be covered in depth in Block 6. Here we introduce them briefly.

Private cloud: these services are **not** provided automatically; you must deploy and manage them yourself (e.g. HAProxy for load balancing, OpenVPN for VPN).

Load Balancer

- Distributes incoming traffic across **multiple instances** (VMs or containers)



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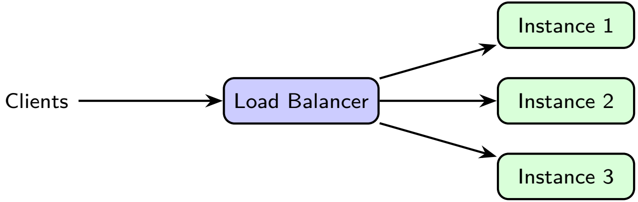

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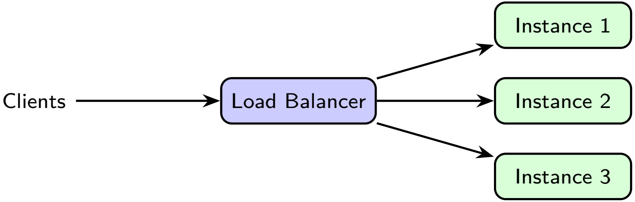

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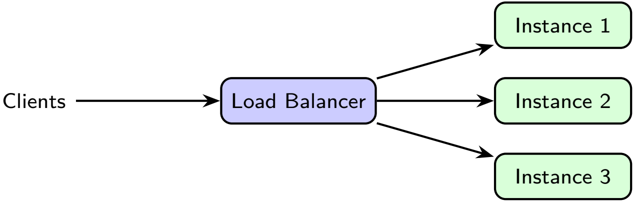

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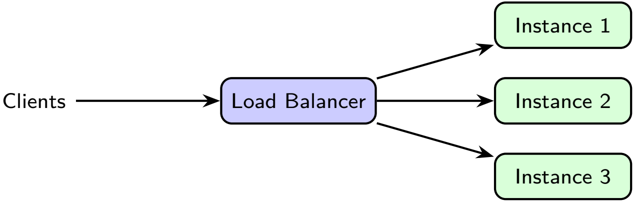

- **L4 (Transport Layer)** Load Balancer: routes based on IP and port (fast, simple)
- **L7 (Application Layer)** Load Balancer: routes based on HTTP headers, URL path, cookies (smarter)
- Cloud examples: AWS ALB/NLB, Azure Load Balancer, GCP Cloud Load Balancing

5

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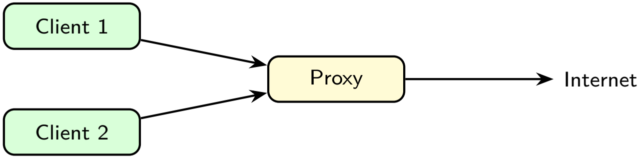

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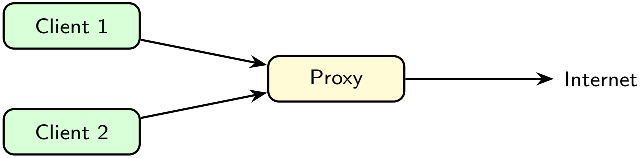

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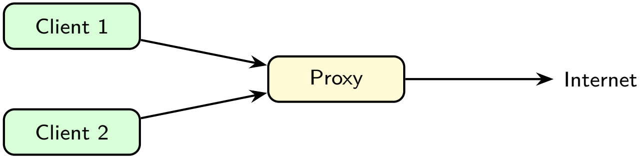

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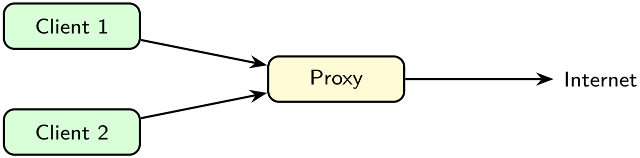

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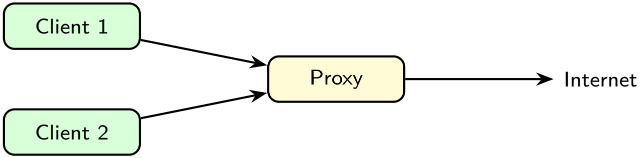

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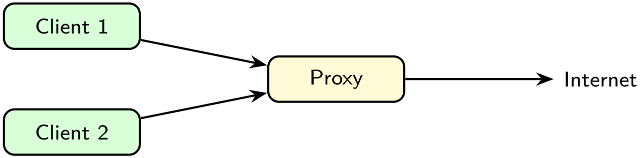

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- Examples: Squid, corporate firewalls with proxy mode

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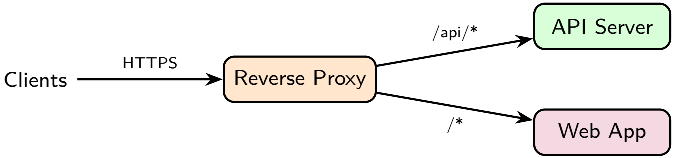

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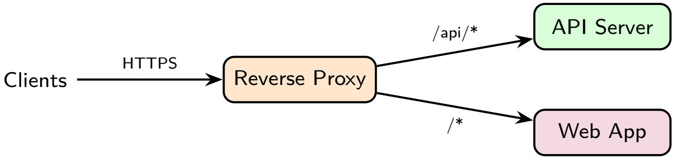

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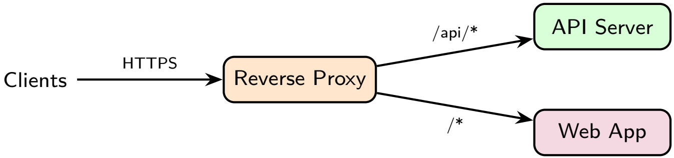

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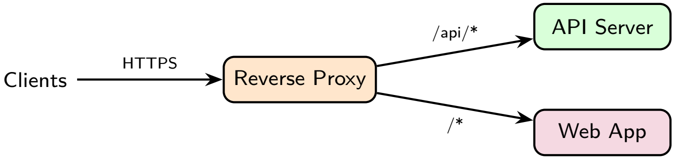

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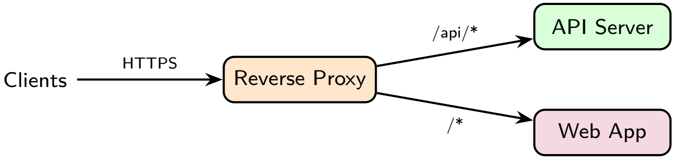

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- Examples: NGINX, HAProxy, AWS CloudFront, Azure Front Door

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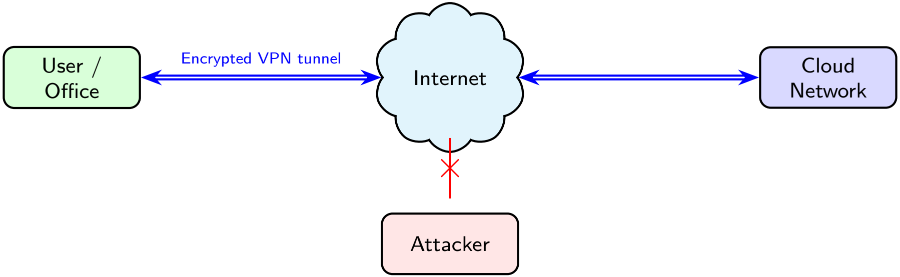
VPN: Virtual Private Network

- A **VPN (Virtual Private Network)** creates a **secure, encrypted tunnel** over the public Internet
- Connects remote networks or users as if they were on the same private network



Two main types:

- **Site-to-site VPN**: office ↔ cloud network
- **Client VPN**: one user ↔ remote network
- Examples: AWS VPN, Azure VPN Gateway, GCP Cloud VPN

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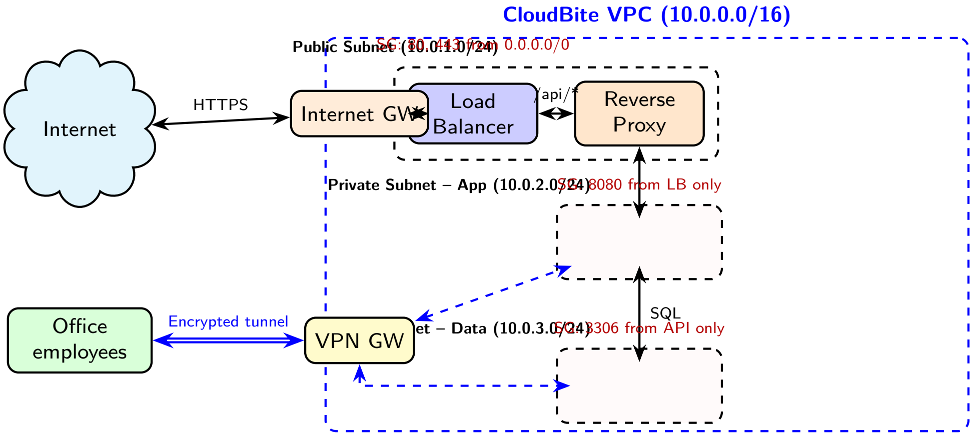
- Security groups + ACLs at each tier

CloudBite Architecture: Putting It All Together



Flow: Users hit the **Load Balancer** (public IP) → **Reverse Proxy** routes /api/* to the **API Server** (private) → API queries the **Database** (private). Office employees access internal resources via the **VPN tunnel**, never traversing the public Internet.

Outline

CloudBite, Five Years Later

- We grew to 2 million users. AWS bill: **$40,000/month** and rising
- Egress fees alone: $5,000/month (data leaving AWS to users)
- The team now has 30 engineers; we *could* run our own servers
- We now ask: "Should we leave the cloud?"

The cloud is not always the cheapest option. Let us explore why.

Case Study: Basecamp Leaves the Cloud

- **Basecamp / 37signals** (creators of Ruby on Rails) ran on AWS for 15 years
- In 2023, they **moved everything to their own hardware**: no new staff needed
- Result: **$10 million saved over five years** (50%+ cost reduction)

[6] 37signals, "We left the cloud," basecamp.com/cloud-exit, 2023.

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Why it worked for them:

- **Stable, predictable workload**: no sudden traffic spikes
- **Large, experienced ops team**: already managing servers regardless
- **No need for global presence**: most users in the same region

Key insight

Basecamp's exit worked because their workload was **predictable**. For a startup with unpredictable traffic, the cloud's elasticity remains essential.

[6]

[6] 37signals, "We left the cloud," basecamp.com/cloud-exit, 2023.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
  - AWS Lambda, DynamoDB, SQS $\rightarrow$ no direct equivalent on Azure or GCP
  - Rewriting the application can cost more than years of cloud bills

[7] Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

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  - AWS charges \$0.09/GB for outbound traffic
  - Moving 50 TB out of AWS $\approx$ \$4,500 in fees alone
  - Egress fees increased 20% in 2023; inter-Availability-Zone (AZ) fees doubled in 2025

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- **68% of enterprises** exceeded their cloud budget due to unexpected data transfer costs (Flexera, 2024)

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The Pricing Ratchet

- Cloud providers attract customers with **low initial pricing and free tiers**
- Once adoption deepens, prices increase:
  - AWS EC2 instance families: significant price rises in 2025
  - Azure: 5%+ on subscriptions, 10% on Premium SSDs (2025)
  - Google Workspace: 20%–34% increase (March 2025)

[8] InfotechLead, "AWS, Azure, and Google Cloud adjust billing models in 2025," 2025.

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  - Google Workspace: 20%–34% increase (March 2025)
- This pattern appears across industries:
  - **Video streaming**: low subscription prices to gain users, then steady increases once the audience is locked in
  - **Cloud computing**: same dynamic, but switching costs are higher because your entire architecture depends on the provider

The pattern

Low prices → adoption → dependency → price increases → switching is too expensive.

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[8] InfotechLead, "AWS, Azure, and Google Cloud adjust billing models in 2025," 2025.

So... Cloud or Not?

	Cloud (public)	On-premises / private
Best when	Unpredictable traffic, small team	Stable workload, large ops team
Cost model	OpEx (pay-as-you-go)	CapEx (buy hardware upfront)
Scaling	Instant, global	Weeks to months (buy + install)
Risk	Vendor lock-in, price hikes	Hardware failures, staffing
Control	Limited (provider's rules)	Full (your hardware, your rules)

- **Most real-world deployments are hybrid**: critical/stable workloads on-premises, burst/variable workloads in the cloud
- There is no universal answer; **every case is a trade-off**

Discussion: The Cloud Trade-Off

CloudBite now has 30 engineers, 2M users, and stable traffic. Should we leave the cloud, go hybrid, or stay? Why?

Think about:

- Is the workload predictable or bursty?
- Does the team have ops expertise?
- How much vendor-specific code exists (Lambda, DynamoDB)?
- What would migration cost in time and money?
- Could a hybrid approach capture the best of both?

Outline

Key Takeaways: CloudBite's Journey

What we built:

1. Virtualization → cloud: technology vs **business model**
2. Five NIST characteristics define true cloud
3. **IaaS / PaaS / SaaS**: who manages what
4. Chose a **platform** (AWS, public cloud)
5. Created a **virtual network** with public and private subnets
6. Secured it with **security groups** + **ACLs**
7. Added **load balancer**, **reverse proxy**, **VPN**

What we learned:

8. The cloud is **not free**: costs grow with scale
9. **Vendor lock-in** makes leaving expensive
10. Prices increase once you depend on the provider
11. On-premises works for **stable, predictable** workloads
12. **Hybrid** is often the pragmatic answer
13. **Every case is a trade-off**; there is no universal solution

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